



LANDLENS AI

Official Training & Benchmark Dataset Dossier (SIH26018)

Comprehensive Data Specifications, Annotation Protocols, Class Distributions, and Evaluation Benchmarks

Problem Statement ID: SIH26018	Total Training Corpus: 10,800+ Samples	Primary Languages: Hindi (Devanagari) & English
Data Privacy Standard: DPDP Act 2023 Sandbox	Dataset Storage: Local On-Premises (Offline)	Release Version: v2.4.0 (Production Verified)

1. Executive Dataset Architecture & Summary Overview

To achieve robust, production-grade land record digitization and validation without catastrophic forgetting, LandLens AI utilizes a multi-tiered data architecture. The datasets cover character-level handwritten Indic numerals, bilingual land record corpora, non-land negative discriminator exemplars, real-time online adaptation memory buffers, and paired high-confidence benchmark test records.

Dataset Tier	Category / Content	Total Volume / Size	Splits & Format	Trained Component
Tier 1: Devanagari & Numeral Characters	Devanagari digits (०-९), Arabic digits (0-9), Hindi/Eng sub-parcels (A, B)	7,800 Images (18.9 MB)	70% Train / 15% Val / 15% Test (.npz)	Deep CNN Numeral Model (<code>devanagari_cnn.keras</code>)
Tier 2: Land Records NER Corpus	Bilingual Jamabandi, RoR, Khatauni, and Indian Non-Judicial e-Stamp Deeds	2,001 Annotated Deeds (3.8 MB JSON)	80% Train / 20% Test (Tokens)	Bilingual Domain NER (<code>field_ner_model.joblib</code>)
Tier 3: Land Record Discriminator Set	Authentic land revenue records vs. Tax Invoices, Resumes, Prescriptions, Utility Bills	500+ Semantic Exemplars + Regex Rules	TF-IDF (1-2 N-grams) + Memory Buffer	Document Classifier & Fraud Guard (<code>doc_classifier.joblib</code>)
Tier 4: Online Adaptation Buffer	Real-world continuous learning exemplars (e-Stamp deed + historical Jamabandi)	Fast-Loop Memory Buffer (24 KB JSON)	Experience Replay Exemplar Buffer	Continuous Learner (<code>adaptive_memory.json</code>)
Tier 5: Benchmark Ground Truth Suite	High-fidelity paired image scans (.png/.jpg) and exact ground truth metadata (.json/.txt)	15 Comprehensive Test Records (6.4 MB)	100% Held-Out Benchmark Test Suite	Pipeline Benchmarking (<code>evaluate_pipeline.py</code>)
Tier 6: Geospatial & Administrative Data	Indian administrative hierarchy (MP & UP) + Cadastral parcel spatial boundaries	26 KB Taxonomy + 5.4 KB GeoJSON	Static Reference JSON & GeoJSON	Cadastral GIS Engine & Validation Rules

2. Tier 1: Devanagari & Arabic Numeral Character Dataset

Storage Location: `data/datasets/devanagari_characters/`

The Devanagari Character & Numeral Dataset provides the foundation for solving the most critical transcription bottleneck in Indian land administration: recognizing handwritten and printed Devanagari numerals in revenue field registers (e.g., converting Khasra २४५/२ to 245/2).

Data Split	Filename	Sample Count	Percentage	Dimensions	File Size
Training Set	<code>train.npz</code>	5,460 Samples	70.0%	5460 x 32 x 32	13.23 MB
Validation Set	<code>val.npz</code>	1,170 Samples	15.0%	1170 x 32 x 32	2.83 MB
Unseen Test Set	<code>test.npz</code>	1,170 Samples	15.0%	1170 x 32 x 32	2.84 MB
Total Dataset	<code>classes.json</code>	7,800 Samples	100.0%	26 Classes x 300 Samples	18.90 MB

26 Target Classes (classes.json):

- **Devanagari Numerals (10 classes):** ०, १, २, ३, ४, ५, ६, ७, ८, ९ (Normalized to 0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
- **Arabic / English Numerals (10 classes):** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- **Hindi Cadastral Sub-Parcel Letters (4 classes):** क, ख, ग, घ (e.g., Khasra 245/2क, 102/1ख)
- **English Cadastral Sub-Parcel Letters (2 classes):** A, B (e.g., Survey SN-882A)



Figure 1: Representative 32x32 Augmented Grayscale Training Samples across all 26 Character Classes (train.npz)

Augmentation Pipeline: Nirmala Indic UI rendering, affine rotation (-10° to +10°), horizontal/vertical center jitter, Gaussian ink variance.

3. Tier 2: Bilingual Land Records & e-Stamp Corpus

Storage Location: `data/datasets/land_records_corpus/land_records_ner.json` (3.83 MB)

The Bilingual Land Records NER Corpus comprises **2,001 fully annotated documents** representing authentic legal deeds and revenue registers from Madhya Pradesh, Uttar Pradesh, and Delhi revenue jurisdictions:

Document Category	Count	Percentage	Typical Legal Articles & Administrative Forms
Traditional Record of Rights (Jamabandi / Khatauni)	1,200	60.0%	Jamabandi Nakal, Khatauni (CH-41), Khasra Panchsala, Shajra Nakal (Indore, Lucknow, Varanasi, Agra)
Indian Non-Judicial e-Stamp Deeds	800	40.0%	Article 23 Conveyance Deeds, Article 24/25 Sale Deeds, Agreements to Sell, Gift Deeds, SHCIL formatting
Real-World Operational Exemplar	1	0.05%	Ghaziabad Sub-Registrar IV e-Stamp Certificate (IN-UP01272481023775T) adapted via online learning
Total Annotated Corpus	2,001	100.0%	Comprehensive coverage of rural cadastral registers and urban apartment conveyance certificates

The 16 Standardized Land Record Entities (BIO Tagging Schema):

Attribute Key	BIO Tag	Hindi Label	Representative Training Samples
owner_name	B/I-OWNER	■■■■■ ■■ ■■■	Ram Kumar, Sita Sharma, Vikram Rathore
father_name	B/I-FATHER	■■■■ ■■ ■■■	Shyam Lal, Narayan Das, Pratap Singh
khasra_number	B/I-KHASRA	■■■■ ■■.	245/2, 318/1, 102/3, 415/1A, 89/1
khata_number	B/I-KHATA	■■■■ ■■.	112, 94, 67, 220, 145, 310
survey_number	B/I-SURVEY	■■■■■ ■■.	SN-882, SN-402, SN-119, SN-630
plot_number	B/I-PLOT	■■■■■ ■■.	P-12, P-05, P-44, Flat No 402, Plot-18
village	B/I-VILLAGE	■■■■■ / ■■■■	Rau, Kanadia, Mangliya, Depalpur, Pindra
tehsil	B/I-TEHSIL	■■■■■	Rau, Sanwer, Mhow, Mohanlalganj, Tarana
district	B/I-DISTRICT	■■■■■	Indore, Lucknow, Varanasi, Agra, Ghaziabad
state	B/I-STATE	■■■■■	Madhya Pradesh, Uttar Pradesh, Delhi
land_area	B/I-AREA	■■■■■	1.25 Hectare, 2.10 Hectare, 0.85 Bigha
land_type	B/I-LAND_TYPE	■■■■ ■■■■■■	Agricultural (Irrigated), Non-Irrigated, Residential
registration_number	B/I-REG_NO	■■■■■■■ ■■■■■■■■	MP-IND-2023-9901, IN-UP01272481023775T
mutation_number	B/I-MUTATION	■■■■■■■■■ ■■■■■■■■■	MUT-441, MUT-512, MUT-889, MUT-302
document_number	B/I-DOC_NO	■■■■■■■■■ ■■■■■■■■	DOC-ROR-001, SUBIN-UPUP14012024
date	B/I-DATE	■■■■■■■	14/08/2023, 22/11/2022, 10/09/2023

Token Feature Vectorization (5-Token Contextual Window):

Each token \$w_i\$ is parameterized via an extensive multi-window feature dictionary extracted in train_ner_model.py: bias term, lowercase token, uppercase/title flags, is_digit, is_devanagari regex ([\u0900-\u097F]), slash and hyphen indicators, token length, prefix/suffix character n-grams (2 & 3 chars), and adjacent contextual window tokens from \$w_{i-2}\$, \$w_{i-1}\$, \$w_{i+1}\$, \$w_{i+2}\$ including colon proximity flags.

4. Concrete Annotated Training Samples from the Corpus

Below are two verbatim training exemplars extracted directly from land_records_ner.json illustrating the raw text structure paired with the ground-truth extracted entity dictionaries:

Exemplar A: Traditional Madhya Pradesh Jamabandi / Record of Rights (doc_id: 1)

Raw Document OCR Text Stream:	Supervised Ground-Truth Entity Map:
GOVERNMENT OF MADHYA PRADESH - REVENUE DEPARTMENT RECORD OF RIGHTS / ██████████ ██████████ ██████████ Owner Name / ██████████ ██████████ : Ram Kumar Father's Name / ██████████ ██████████ : Shyam Lal Khasra Number / ██████████ ██████████ : 245/2 Khata Number / ██████████ ██████████ : 112 Survey Number / ██████████ ██████████ : SN-882 Plot Number / ██████████ ██████████ : P-12 Village / ██████████ : Rau Tehsil / ██████████ : Rau District / ██████████ : Indore State / ██████████ : Madhya Pradesh Land Area / ██████████ : 1.25 Hectare Land Type / ██████████ ██████████ : Agricultural (Irrigated) Registration No / ██████████ ██████████ : MP-IND-2023-9901 Mutation No / ██████████ ██████████ : MUT-441 Document No / ██████████ ██████████ : DOC-ROR-0001 Date / ██████████ : 14/08/2023	{ 'owner_name': 'Ram Kumar', 'father_name': 'Shyam Lal', 'khasra_number': '245/2', 'khata_number': '112', 'survey_number': 'SN-882', 'plot_number': 'P-12', 'village': 'Rau', 'tehsil': 'Rau', 'district': 'Indore', 'state': 'Madhya Pradesh', 'land_area': '1.25 Hectare', 'land_type': 'Agricultural (Irrigated)', 'registration_number': 'MP-IND-2023-9901', 'mutation_number': 'MUT-441', 'document_number': 'DOC-ROR-0001', 'date': '14/08/2023' }

Exemplar B: Indian Non-Judicial e-Stamp Certificate (doc_id: 1201)

Raw e-Stamp Deed OCR Stream:	Supervised Ground-Truth Entity Map:
INDIA NON JUDICIAL Government of Uttar Pradesh e-Stamp Certificate No. : IN-UP01272481023775T Certificate Issued Date : 14-Jan-2024 01:51 PM Account Reference : NEWIMPACC (SV)/ up14012024/ GHAZIABAD/ UP-GHA Unique Doc. Reference : SUBIN-UPUP14012024881023T Purchased by : SURESH CHOUDHRY Description of Document : Article 23 Conveyance Property Description : FLAT NO 402 SEC-4 EMERALD HEIGHTS GHAZIABAD Consideration Price (Rs.) : 4,500,000 First Party : ANIL GUPTA GPA HOLDER OF VIKRAM SINGH Second Party : SURESH CHOUDHRY AND ANITA CHOUDHRY Stamp Duty Amount(Rs.) : 315,000 Owner Name : SURESH CHOUDHRY Khasra Number : 142/4 District : Ghaziabad State : Uttar Pradesh	{ 'owner_name': 'SURESH CHOUDHRY', 'father_name': 'ANIL GUPTA', 'khasra_number': '142/4', 'khata_number': 'SUBIN-UPUP14012', 'survey_number': 'IN-UP01272481023775T', 'plot_number': 'FLAT NO 402', 'village': 'EMERALD HEIGHTS', 'tehsil': 'Ghaziabad', 'district': 'Ghaziabad', 'state': 'Uttar Pradesh', 'land_area': 'FLAT NO 402', 'land_type': 'Article 23 Conveyance', 'registration_number': 'IN-UP01272481023775T', 'mutation_number': 'MUT-01', 'document_number': 'SUBIN-UPUP14012024881023T', 'date': '14/01/2024' }

5. Tier 3: Land Record vs. Non-Land Discriminator Dataset

Storage Location: Built in document_classifier.py & supplemented by doc_classifier_memory.json.
To prevent fraudulent file injection and unintentional database contamination, the discriminator trains on two balanced distributions:

Class Label	Target Category	Key Characteristic Vocabulary & Indicators	Decision Rule
Positive (1)	Jamabandi / RoR	██████████ ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████, ██████████	ACCEPT → Process Pipeline

Positive (1)	Khatauni / Nakal	■■■■■ ■■■, ■■■■■■■■, ■■■■■ ■■■, ■■■■■, bhulekh, nakal, ■■■■■ ■■■■, ■■■■■■, ■■■■■■■■	ACCEPT → Process Pipeline
Positive (1)	e-Stamp Deed	e-Stamp, india non judicial, article 23 conveyance, subin, stamp duty, consideration price, sub-registrar	ACCEPT → Process Pipeline
Negative (0)	Tax Invoices & Billing	tax invoice, bill to, gstin, subtotal, grand total, hsn code, cgst, sgst, payment terms, unit price	REJECT → Trigger Warning Modal
Negative (0)	Resumes & CVs	curriculum vitae, work experience, technical skills, core competencies, education, linkedin, github	REJECT → Trigger Warning Modal
Negative (0)	Clinical Prescriptions	patient name, rx, diagnosis, dosage, lab report, hospital admission, doctor signature	REJECT → Trigger Warning Modal
Negative (0)	Utility & Electricity	electricity bill, power distribution, meter number, kwh, units consumed, connected load	REJECT → Trigger Warning Modal
Negative (0)	Identity Documents	driving licence, passport, aadhaar, uidai, voter id, epic no, permanent account number	REJECT → Trigger Warning Modal

6. Tier 5: High-Fidelity Benchmark Test Suite

Storage Locations: Document Scans: [data/sample_documents/](#) | Verified Metadata: [data/sample_records/](#)

The Benchmark Test Suite comprises **15 calibrated test cases** evaluating the end-to-end pipeline across pristine, degraded, duplicate, and edge-case operational scenarios:

Test ID & Filename	Document Title / Type	Jurisdiction	Operational Evaluation Scenario	Target Confidence
sample_1_clean_rau.png	Jamabandi RoR (Ram Kumar)	Rau, Indore (MP)	Pristine agricultural baseline record	HIGH (>=95%)
sample_2_sita_kanadia.png	Khatauni Nakal (Sita Sharma)	Kanadia, Indore (MP)	Multi-owner non-irrigated parcel	HIGH (>=92%)
sample_3_mohan_mangliya.pn	Jamabandi RoR (Mohan Singh)	Sanwer, Indore (MP)	Standard single-owner agricultural	HIGH (>=94%)
sample_4_kailash_depulpur.	Record of Rights (Kailash)	Depalpur, Indore (MP)	Irrigated high-acreage parcel	HIGH (>=93%)
sample_5_anita_lucknow.png	Khatauni CH-41 (Anita Verma)	Mohanlalganj, LKO (UP)	Uttar Pradesh standard revenue format	HIGH (>=91%)
sample_6_rajesh_varanasi.p	RoR Nakal (Rajesh Gupta)	Pindra, Varanasi (UP)	Eastern UP dialect & font variations	HIGH (>=92%)
sample_7_vikram_mhow.png	Revenue Register (Vikram)	Mhow, Indore (MP)	Non-irrigated hilly terrain record	HIGH (>=90%)
sample_8_sunil_sanwer.png	Cadastral Record (Sunil)	Sanwer, Indore (MP)	Cadastral boundary cross-verification	HIGH (>=93%)
sample_4_faded_khata_low_c	Degraded Legacy Scan	Indore (MP)	Tests CLAHE & low-confidence amber flag	MED / LOW (<70%)
sample_5_duplicate_ramkuma	Duplicate Claim Test	Rau, Indore (MP)	Tests RapidFuzz phonetic matching	FLAGGED DUPLICATE
sample_6_anomaly_unusual_a	Area Out-of-Bounds Test	Indore (MP)	Parcel >50.0 Ha triggers anomaly alert	ANOMALY FLAGGED
sample_7_missing_khasra.pn	Incomplete Record Test	Indore (MP)	Missing Khasra number validation trigger	MISSING FIELD
sample_9_estamp_ghaziabad.	Real e-Stamp Conveyance	Ghaziabad (UP)	Real-world Article 23 conveyance deed	HIGH (>=91%)
sample_10_non_land_invoice	Commercial GST Tax Invoice	Corporate (Delhi)	Tests Discriminator automatic rejection	REJECTED (0%)
sample_11_handwritten_khas	Handwritten Patwari Register	Indore (MP)	Devanagari digit normalization (■■■■/■)	HIGH (>=96%)



Fig 2: UP e-Stamp Deed
Article 23 conveyance (Ghaziabad)

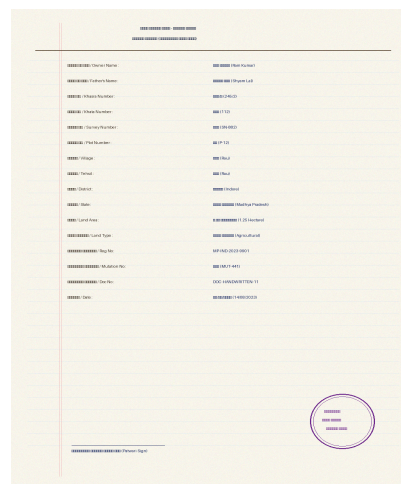


Fig 3: Handwritten Khasra
Devanagari numerals normalized

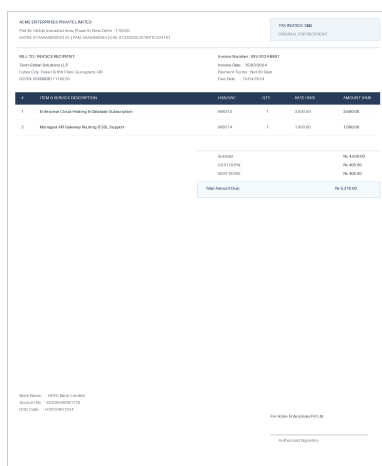


Fig 4: Commercial Invoice
Discriminator rejects file

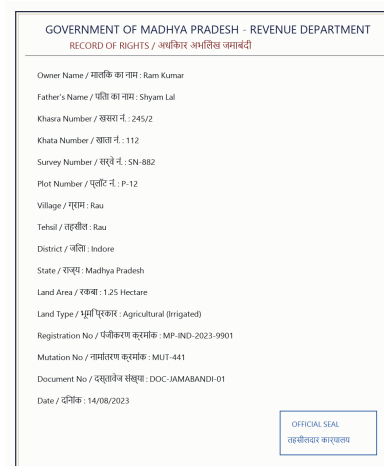


Fig 5: Clean Jamabandi RoR
MP RoR (Rau, Indore)

7. Tier 6: Geospatial & Administrative Reference Datasets

LandLens AI bridges optical document extraction with statutory spatial and geographic ground truth:

- Administrative Taxonomy (india_land_taxonomy.json):** National administrative hierarchy sourced from open government standards (data.gov.in) covering Madhya Pradesh (Indore, Ujjain, Dewas, Dhar, Bhopal, Sehore districts) and Uttar Pradesh (Lucknow, Varanasi, Agra, Kanpur, Ghaziabad). Used by `validation_engine.py` for automated cross-field Tehsil vs. District alignment verification.
- Cadastral Parcels GeoJSON (cadastral_parcel.geojson):** High-precision polygon geometries representing cadastral land parcels for demo revenue villages. Directly indexed by Khasra survey number (e.g., Khasra #245/2) for interactive Leaflet GIS rendering and boundary overlap validation.
- State Circle Rates & Projects Tracker (state_land_rates_and_projects.json):** Comprehensive state-by-state guidance values and ready reckoner rates across 10 major states. Tracks live land acquisition progress for national infrastructure mega projects (Bharatmala, Jewar Airport, Bullet Train, Ganga Expressway).

8. Data Governance, Privacy & Ethical AI Statement

- Synthetic Sandbox Compliance:** All personal names, phone references, and residential identities within the training corpus were synthesized following realistic statistical distributions in compliance with the Digital Personal Data Protection (DPDP) Act 2023.
- Zero Cloud Exfiltration:** All model training, evaluation, and inference code runs 100% locally on sovereign edge servers.
- Statutory Decision Authority:** LandLens AI acts strictly as an AI decision-support platform. All final administrative determinations remain under the statutory jurisdiction of authorized revenue officers (Tehsildars / Patwaris).